CowMoo

A 3D Digital twin engine for virtual cattle health

Project Member

SAN MEYCHING, [6026010018], [ms6026010018@camtech.edu.kh]

PRITHY MONYNICH, [6026010076], [mp6026010076@camtech.edu.kh]

PICH SOKSOVANN, [6026010074], [sp6026010074@camtech.edu.kh]

BIN PHEAKVIREAKBOT, [6026010111], [pb6026010111@camtech.edu.kh]

TOCH VANLYVEY, [6026010182], [vt6026010182@camtech.edu.kh]

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Academic Year: Year I

Academic Term: Term III

SUPERVISOR: VONEAT PEN

CO-SUPERVISOR: LIPHENG PRUM

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I. Introduction

- **Problem Statement:** Farmers and livestock researchers do not have an easy way to run "What if" scenarios before making change in the barn. Relying only on live physical trials limits how many diets or management strategies can be tested, slows down research on reducing greenhouse gas emissions, and risks putting unnecessary stress on animals.
- **Competitive Landscape:** Today's livestock tech is great at logging live, daily activity, but there is a clear gap when it comes to forward looking biological simulation. The academic modeling tools that do exist are mostly text heavy spreadsheets that are difficult to interpret and hard for non-engineering to use quickly.

II. Objectives

- **The Solution:** We are building a software base 3D digital twins of cows using open source simulation frameworks and godot rendering engine. The virtual cow will process inputs like ambient temperature and feed quality, displaying real time visual indicators and predict for heat stress, metabolic outputs, health , and energy balance.
- **Impact & Value Proposition:** CowMoo provides a risk free digital environment for agriculture testing. Farmers and livestock researchers can experiment without endangering herd health or wasting resources on live trails. It acts as a digital sandbox for farm management, allowing users to confidently run the scenarios and optimize their strategies entirely on a computer before making any actual change in the barn.
- **Feasibility & Scalability:** This project is highly feasible it relies exclusively on free, open source

III. Proposed Solutions & Methodology

How the system work (4-Step Flow)

1. **Input data**: The User enters basic information like air temperature, humidity, and feed quality or amount.
2. **Calculate Health:** A Python program uses animal science formulas (THI : Temperature Humanity index) to figure out the cow's heat stress, energy levels, and body condition.
3. **Show 3D Visuals:** A 3D cow model made in the Godot engine updates live on screen (for example, the 3D cow turning red when it gets too hot).
4. **Save Results**: The system saves every test scenario so users can compare different situations side by side over time.

Technology Stack & Justifications

1. **3D Graphics Engine: Godot Engine**

   Justification: It is completely free, open source, and easy for students to learn while still making realistic 3D cow models.

2. **Health Calculations: Python**

   Justification: It is the best programming language for science and math, and our data team already knows how to use it.

3. **Website Interface: React.js & Tailwind CSS**

   Justification: React helps us make a smooth, easy to use dashboard, and Tailwind keeps the site looking clean on both phones and computers.

4. **Data Storage & Web Hosting: Firebase & Vercel**

   Justification: Both offer free plans for student projects, require no complex server setups, and let us put website updates online fast.

Development Methodology

Agile Development: it helps us build the project step by step. We can test small pieces as we go and fix things quickly, instead of building everything all at once and hoping it works at the end.

IV. Scope and Limitations

In-Scope:

- A 3D virtual cow model in Godot that reacts visually to temperature and feed adjustments.
- A health calculation engine tracking three core metrics: heat stress, feed/energy balance, and basic metabolic rate.
- A web dashboard with input sliders so users can adjust settings (like room temperature) and watch the cow respond in real time.
- A built in history log that saves past simulation runs for side by side comparison.

Out-of-Scope:

- Live camera feeds or physical hardware installed on real farms (planned for future project phases beyond Year 1).
- Multispecies support like pigs or chickens (the Year 1 build focuses strictly on cows).

- Medical diagnostics or veterinary treatment advice (this tool is strictly for simulation, research, and educational awareness).
- Native mobile app store releases (we are delivering a fully responsive web dashboard instead).

V. Expected Outcomes & Deliverables

By the end of Term III, the following will be completed:

1. Live Web Prototype: A working website link where anyone can open and test the interactive 3D cow model in real time.
2. Complete Source Code: A clean, well explained GitHub repository containing all our code from the Python health formulas to the Godot 3D graphics and React web dashboard.
3. System Architecture Diagram: A clear visual diagram showing how data moves step by step from user inputs, through calculations, and onto the 3D screen.
4. User Guide: A quick, friendly manual explaining how to adjust the dashboard sliders and easily read the cow's visual health signals.
5. Final Project Report: A complete write up detailing our research background, design choices, testing results, and academic references.

VI. Work Plan

Guidelines: Provide a clear timeline for your project. We highly recommend using a table format divided into phases, milestones, or weeks. This shows your mentors that you can plan your time realistically.

Example:

Phase	Weeks	Activities	Deliverables / Milestones
Phase 1: Research & Architecture & Technical Foundations	Weeks 1 - 3	Define user requirements and biological parameters(Temperature humidity index). Set up the Github repository and database structures.	System architecture diagram approved; repository initialized.
Phase 2: Core math engine & data simulation (sprint 1)	Weeks 3 - 6	Develop unit test python algorithms for heat stress entrance and energy balance. Build data generators to simulate mock environmental inputs (temperature, humidity,	Verified python calculation module producing validated numerical outputs.

		feed quality) without hardware dependencies.	
Phase 3: Visualization& dashboard UI (sprint 2)	Weeks 5 - 9	Animate a rigged 3D cow. Build react.js and Tailwind CSS web interface with interactive sliders state management. Create local communication hooks between the UI and rendering engine.	Function 3D cow scene responding to visual triggers; completed frontend dashboard UI
Phase 4: System Integration& Pipeline testing	Weeks 8 - 11	Connect the Python backend to logic to react frontend and Godot 3D review. Set up a firebase to save up each simulation run, allowing to pull up and compare past scenarios anytime. Run full system tests from start to finish to catch pipeline bugs, cut down 3D rendering lag, and make sure the whole app runs smoothly.	Fully connected end-to-end prototype, test log report, stable simulation pipeline.
Phase 5: Deployment & Reporting	Weeks 11 - 12	Deploy the web dashboard to Vercel	Live website link; final presentation.

VII. Risk Management & Mitigation

Guidelines: Every project has risks. Identifying them in advance shows high maturity. Think about what could go wrong (technical, personal, or organizational) and how you will handle it.

Example:

Identified Risk	Impact / Likelihood	Preventive Action / Mitigation
Web Performance & 3D Lag	High / Low	
Engine-to-Frontend Integration Failure	High / Medium	

Formula Inaccuracy & Calibration Errors	High / Medium	

VIII. Budget (Optional)

Guidelines: If your project requires software subscriptions, hardware components, or cloud hosting services that cost money, list them here. If everything you are using is free, state that clearly.

Example:

Because this is an internship project with a focus on cost-efficiency, we are utilizing free tiers of modern cloud software. The project has a **$0 budget footprint**.

- **Firebase (Database & Auth):** Spark Free Tier ($0/month) - sufficient for up to 50,000 reads/day.
- **Vercel (Hosting):** Hobby Tier ($0/month) - sufficient for non-commercial web projects.
- **Development Tools:** VS Code, Git, Figma ($0) - all open-source or free personal tiers.

IX. References

Guidelines: Provide a list of books, academic papers, documentation, or websites that helped you form your proposal. Use a standard academic format like APA or IEEE.

Example:

- **APA Format Examples:**
    - Cooper, A., Reimann, R., & Cronin, D. (2014). *About Face: The Essentials of Interaction Design* (4th ed.). John Wiley & Sons.
    - Firebase. (2026). *Cloud Firestore Documentation*. Google Developers. Retrieved from https://firebase.google.com/docs/firestore
    - Hunt, J., & Thomas, D. (2019). *The Pragmatic Programmer: Your Journey to Mastery* (20th Anniversary Edition). Addison-Wesley Professional.